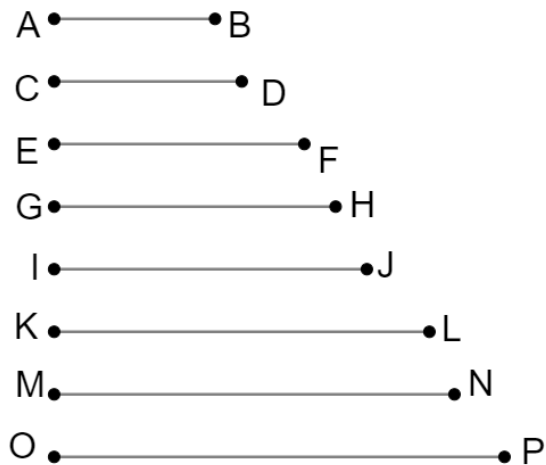
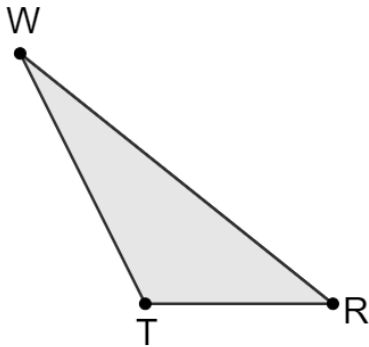


Geometry
Unit 5
Lesson 1 Practice

Name _____
Date _____
Period _____

Triangle WRT and 8 line segments are shown.

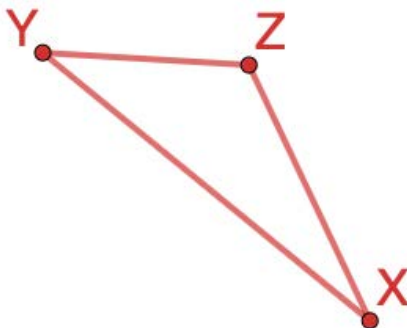


Use patty paper, a ruler, or edge of a piece of paper to determine which line segments have the same measure as each side of triangle WRT .

Complete each.

1. $WT = \underline{\text{GH}}$
2. $RT = \underline{\text{CD}}$
3. $WR = \underline{\text{MN}}$
4. Construct a triangle using the line segments that have the same measure as each side of triangle WRT . Name the new triangle XYZ .

Answers may vary.



5. Complete the statement.

The three sides of triangle WRT are congruent to the three sides of

triangle XYZ because $WT =$

☐	$XY,$
☐	$YZ,$
☐	$XZ,$

 $RT =$

☐	$XY,$
☐	$YZ,$
☐	$XZ,$

 and

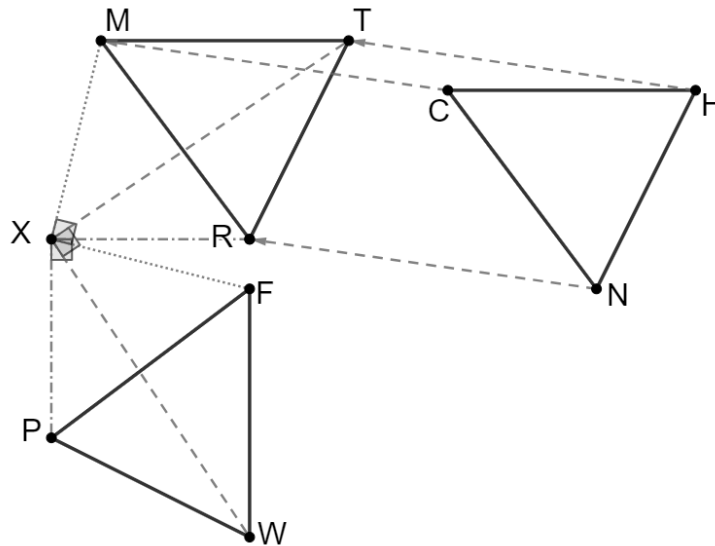
$WR =$

☐	$XY.$
☐	$YZ.$
☐	$XZ.$

6. Determine if the angles of $\triangle WRT$ and $\triangle XYZ$ are congruent. Explain how you determined your answer.

$\triangle WRT \cong \triangle XYZ$ by the Side-Side-Side Congruence Postulate because the three sides of $\triangle WRT$ are congruent to the three sides of $\triangle XYZ$.

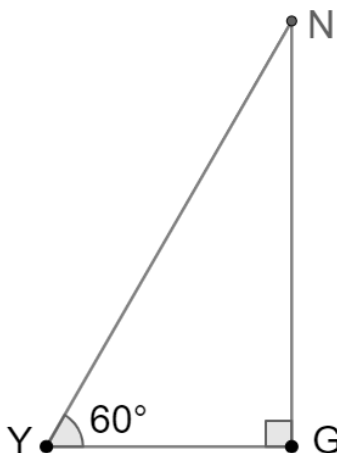
A series of transformations were applied to $\overline{CH}$, $\overline{HN}$, and $\overline{NC}$.



Complete the statements.

7. The first transformation applied to $\overline{CH}$, $\overline{HN}$, and $\overline{NC}$ is a translation. This results in $CH =$ MT , $HN =$ TR and $NC =$ RM .
8. Therefore, by the definition of congruence, $\triangle HNC \cong \triangle TRM$.
9. The second transformation applied to $\overline{MR}$, $\overline{RT}$, and $\overline{TM}$ is a rotation. This results in $MR =$ FP , $RT =$ PW and $TM =$ FW .
10. Therefore, by the definition of congruence, $\triangle TRM \cong \triangle$ WPF .

Triangle NGY is shown where $m\angle Y = 60^\circ$ and $m\angle G = 90^\circ$.



Complete the steps in the construction of $\triangle HRT$ so that $\triangle HRT \cong \triangle NGY$ using two angles of $\triangle NGY$ and their included side.

- | | |
|----|---|
| 1. | Copy <u>$\overline{GY}$</u> to construct $\overline{RT}$ so that <u>$\overline{GY}$</u> $\cong$ $\overline{RT}$. |
| 2. | Copy <u>$\angle Y$</u> to construct $\angle T$ so that <u>$\angle Y$</u> $\cong$ $\angle T$. |
| 3. | Copy <u>$\angle G$</u> to construct $\angle R$ so that <u>$\angle G$</u> $\cong$ $\angle R$. |
| 4. | Label the intersection of the sides of the two angles H . |

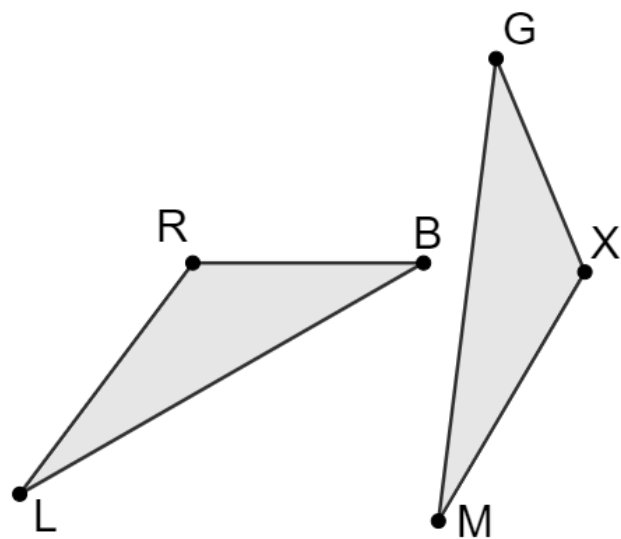
5. Describe how to use geometric tools to determine if $\overline{YN} \cong \overline{TH}$, $\overline{GN} \cong \overline{RH}$, and $\angle N \cong \angle H$.

A straight edge, patty paper or ruler can be used to measure the lengths of each side to determine congruence. A protractor, compass or patty paper can be used to measure each angle and determine congruence.

6. What postulate or theorem does the construction illustrate?

Angle-Side-Angle (ASA) Congruence

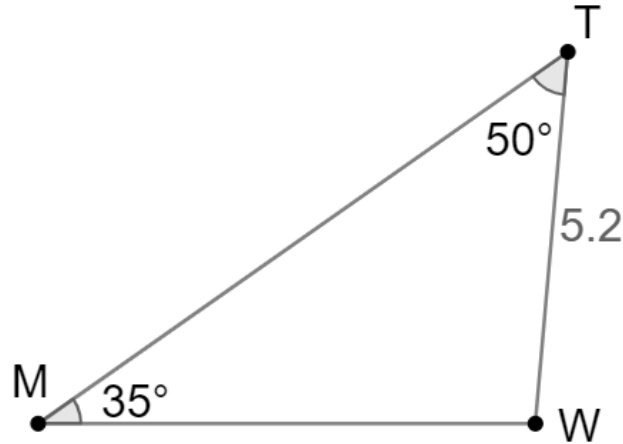
Triangles RBL and GXM are shown.



Identify if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate.

	Given Information	Are the Triangles Congruent?	Theorem or Postulate Used
7.	$\overline{XG} \cong \overline{RB}, \angle RBL \cong \angle XGM, \overline{BL} \cong \overline{GM}$	Yes	SAS
8.	$\angle BLR \cong \angle XMG, \overline{RL} \cong \overline{XM}, \angle BRL \cong \angle GXM$	Yes	ASA
9.	$\overline{BL} \cong \overline{GM}, \angle BLR \cong \angle XMG, \overline{XG} \cong \overline{RB}$	No	
10.	$\overline{XG} \cong \overline{RB}, \overline{BL} \cong \overline{GM}, \overline{RL} \cong \overline{XM}$	Yes	SSS

Triangle TWM is shown where $m\angle M = 35^\circ$, $m\angle T = 50^\circ$, and $WT = 5.2$ units.



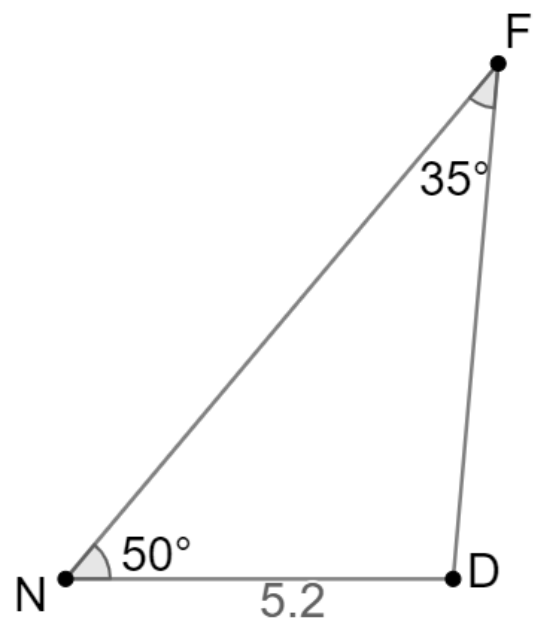
Complete the steps to describe how to construct a triangle that is congruent to $\triangle TWM$ using the angle-angle-side criteria where the first step is to draw a line that is longer than all of the sides of $\triangle TWM$.

1. Use a compass or straightedge to copy TW . Label the endpoints as N and D .
2. Use a compass to copy angle T . Construct a congruent angle using point N as the vertex.
3. Use a compass to copy angle M . Construct a congruent angle so that one side of the angle goes through point D .

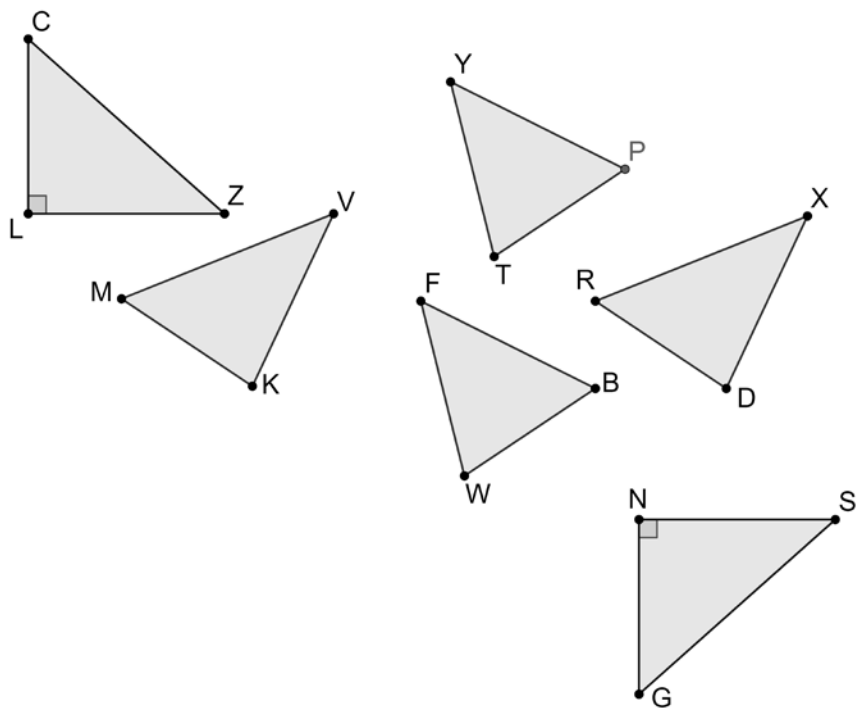
Shanice used the steps to create a triangle that is congruent to $\triangle TWM$. Her triangle is shown.

4. Describe what geometric tools Shanice can use to ensure $\triangle TWM \cong \triangle NDF$.

Shanice can use a ruler, compass, protractor, and/or patty paper.



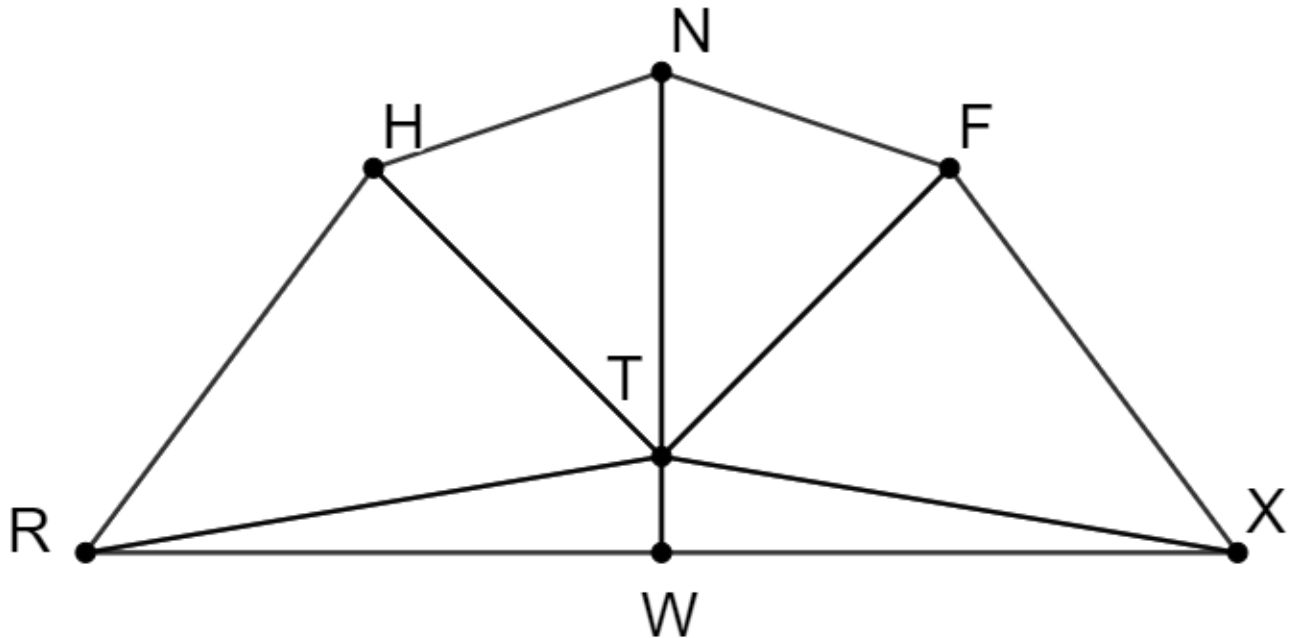
Triangles CLZ , MKV , YPT , BFW , GSN , and XDR are shown.



Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate.

	Given Information	Are the Triangles Congruent?	Theorem or Postulate Used
5.	$\overline{LZ} \cong \overline{NS}$, $\overline{CZ} \cong \overline{SG}$, $\angle LZC \cong \angle NSG$	Yes	SAS Congruence or HL
6.	$\angle PTY \cong \angle BWF$, $\overline{YP} \cong \overline{FB}$, $\angle TYP \cong \angle WFB$	Yes	AAS Congruence
7.	$\overline{VK} \cong \overline{XD}$, $\overline{MK} \cong \overline{DR}$, $\angle KVM \cong \angle DXR$	No	
8.	$\overline{CZ} \cong \overline{GS}$, $\overline{CL} \cong \overline{GN}$	Yes	HL
9.	$\overline{MV} \cong \overline{XR}$, $\overline{KM} \cong \overline{RD}$, $\overline{XD} \cong \overline{VK}$	Yes	SSS Congruence
10.	$\angle VMK \cong \angle XRD$, $\overline{VK} \cong \overline{XD}$, $\overline{MK} \cong \overline{DR}$	No	

Polygon $RHNFX$ is shown.



1. List all of the triangles in the diagram.
 $\triangle RTW$, $\triangle RTX$, $\triangle XTW$, $\triangle TFX$, $\triangle TNF$, $\triangle TNH$, $\triangle THR$

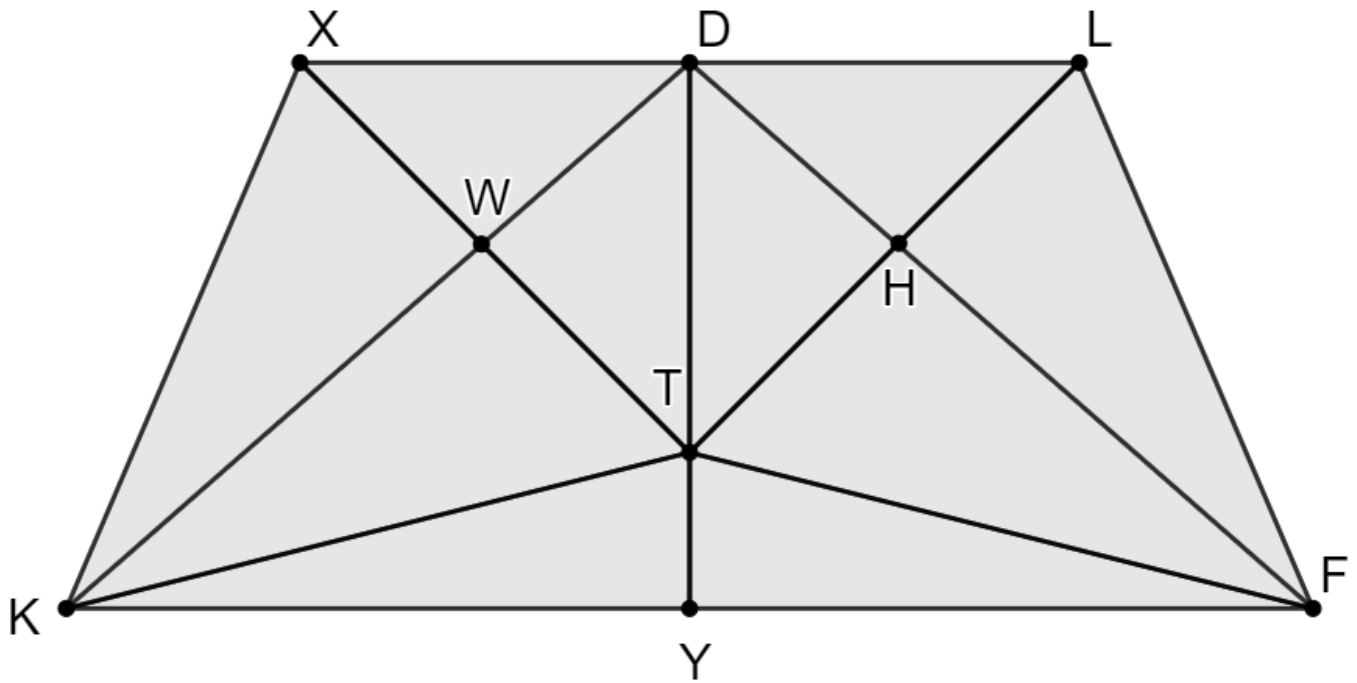
If $\triangle RTH \cong \triangle XTF$, complete the congruence statements.

2. $\overline{TH} \cong \overline{TF}$
3. $\angle TRH \cong \angle TFX$

If $\triangle RTW \cong \triangle XTW$, complete the congruence statement.

4. $\angle TXW \cong \angle TRW$

Quadrilateral $XLFK$ is shown.



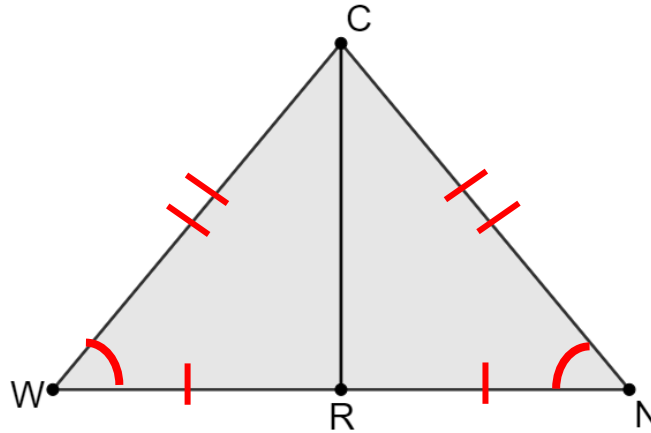
If $\triangle XTD \cong \triangle LTD$, complete the congruence statements.

5. $\overline{XT} \cong \overline{LT}$
6. $\angle DLT \cong \angle DXT$
7. $\overline{DT} \cong \overline{DT}$

If $\triangle KDX \cong \triangle FDL$, complete the congruence statements.

8. $\overline{DF} \cong \overline{DK}$
9. $\angle XDK \cong \angle LDF$
10. $\angle FLD \cong \angle KXD$

Triangle CWN is shown where R is the midpoint of $\overline{WN}$.



1. Triangle $\triangle CWN$ is an isosceles triangle.
Mark the diagram using the given information.
2. Which side(s) of triangles CRW and NRC are congruent because of the given information?

$$\overline{WR} \cong \overline{RN} \text{ and } \overline{CW} \cong \overline{CN}$$

3. Which angle(s) of triangles CRW and NRC are congruent because of the given information?

$$\angle CWR \cong \angle CNR$$

4. Which side(s) or angle(s) of triangles CRW and NRC are congruent because of the reflexive property?

$$\overline{CR} \cong \overline{CR}$$

5. Using the given information, which if any of the triangle congruence theorems or postulates could be used to determine that the triangles are congruent.

☒ SSS

☒ SAS

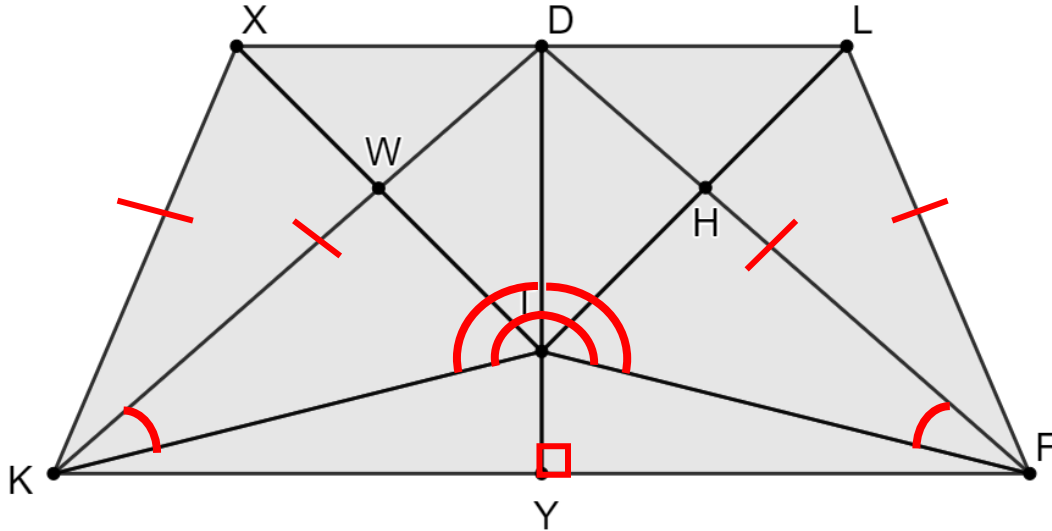
☐ ASA

☐ AAS

☐ HL

☐ none

Quadrilateral $KXLF$ is shown, where $\angle DKT \cong \angle TFH$, $\overline{KD} \cong \overline{DF}$, $\angle DTK \cong \angle FTD$, $\overline{XK} \cong \overline{LF}$, and $\overline{DY}$ is perpendicular to $\overline{KF}$.



6. Mark the diagram with the given information.

7. Complete the statement.

$\triangle_TKD_ \cong \triangle_TFD_$ because $\overline{KD} \cong \overline{DF}$, $\angle DTK \cong \angle FTD$, and $_\angle DKT \cong \angle DFT_$.

8. Which postulate(s) or theorem(s) can be used to show that $\triangle KDY \cong \triangle FDY$?

☐ SSS ☐ SAS ☐ ASA ☐ AAS ☒ HL ☐ none

9. Which postulate(s) or theorem(s) can be used to show that $\triangle DKX \cong \triangle DFL$?

☒ SSS ☒ SAS ☐ ASA ☐ AAS ☐ HL ☐ none

10. Which postulate(s) or theorem(s) can be used to show that $\triangle TDF \cong \triangle TDK$?

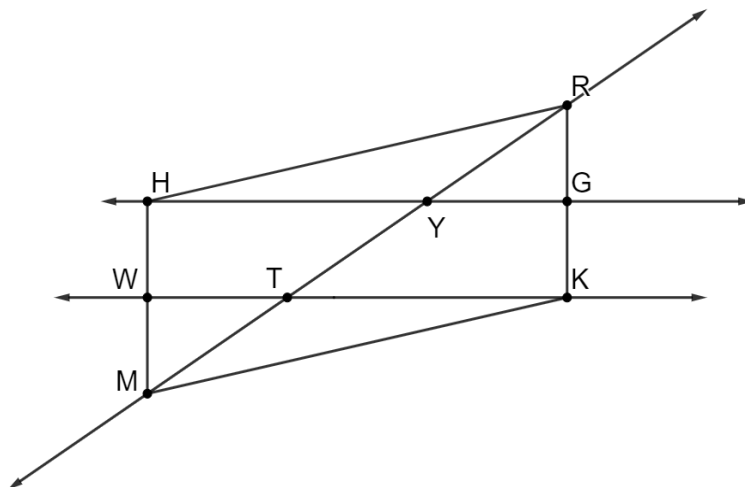
☐ SSS ☐ SAS ☒ ASA ☐ AAS ☐ HL ☐ none

Geometry
Unit 5
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

A diagram is shown where $\overline{HG} \parallel \overline{WK}$,
 $\overline{HR} \parallel \overline{MK}$, and $\overline{HR} \cong \overline{MK}$.

Use the given information to answer
the questions about the incomplete
proof that is shown.

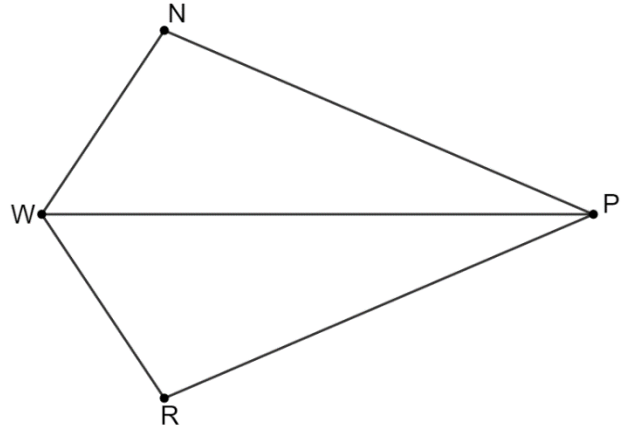


Statement	Reason
1. $\overline{HG} \parallel \overline{WK}$	1. Given
2.	2.
3. $\overline{HR} \parallel \overline{MK}$	3. Given
4.	4.
5. $\overline{HR} \cong \overline{MK}$	5. Given
6. $\triangle HYR \cong \triangle KTM$	6.

- What is the statement for line 2 of the proof?
 $\angle HYR \cong \angle KTM$
- What is the reason for line 2 of the proof?
If two parallel lines are intersected by a transversal, then alternate exterior angles are congruent.
- What is the statement for line 4 of the proof?
 $\angle HRY \cong \angle KMT$
- What is the reason for line 4 of the proof?
If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.
- What is the reason for line 6 of the proof?
AAS Congruence

A quadrilateral is shown where $\overline{WP}$ bisects $\angle NPR$, and $\overline{NP} \cong \overline{PR}$.

Use the given information to answer the questions about the incomplete proof that is shown.



Statement	Reason
1. $\overline{NP} \cong \overline{PR}$	1. Given
2. $\overline{WP}$ bisects $\angle NPR$	2. Given
3.	3.
4.	4. Reflexive property
5. $\triangle PRW \cong \triangle$ _____	5.

6. What is the statement for line 3 of the proof?
 $\angle NPW \cong \angle RPW$

7. What is the reason for line 3 of the proof?
definition of angle bisector

8. What is the statement for line 4 of the proof?
 $\overline{WP} \cong \overline{WP}$

9. Complete the congruence statement for the statement of line 5 of the proof.

$$\triangle PRW \cong \triangle PNW$$

10. What is the reason for line 5 of the proof?
SAS Congruence

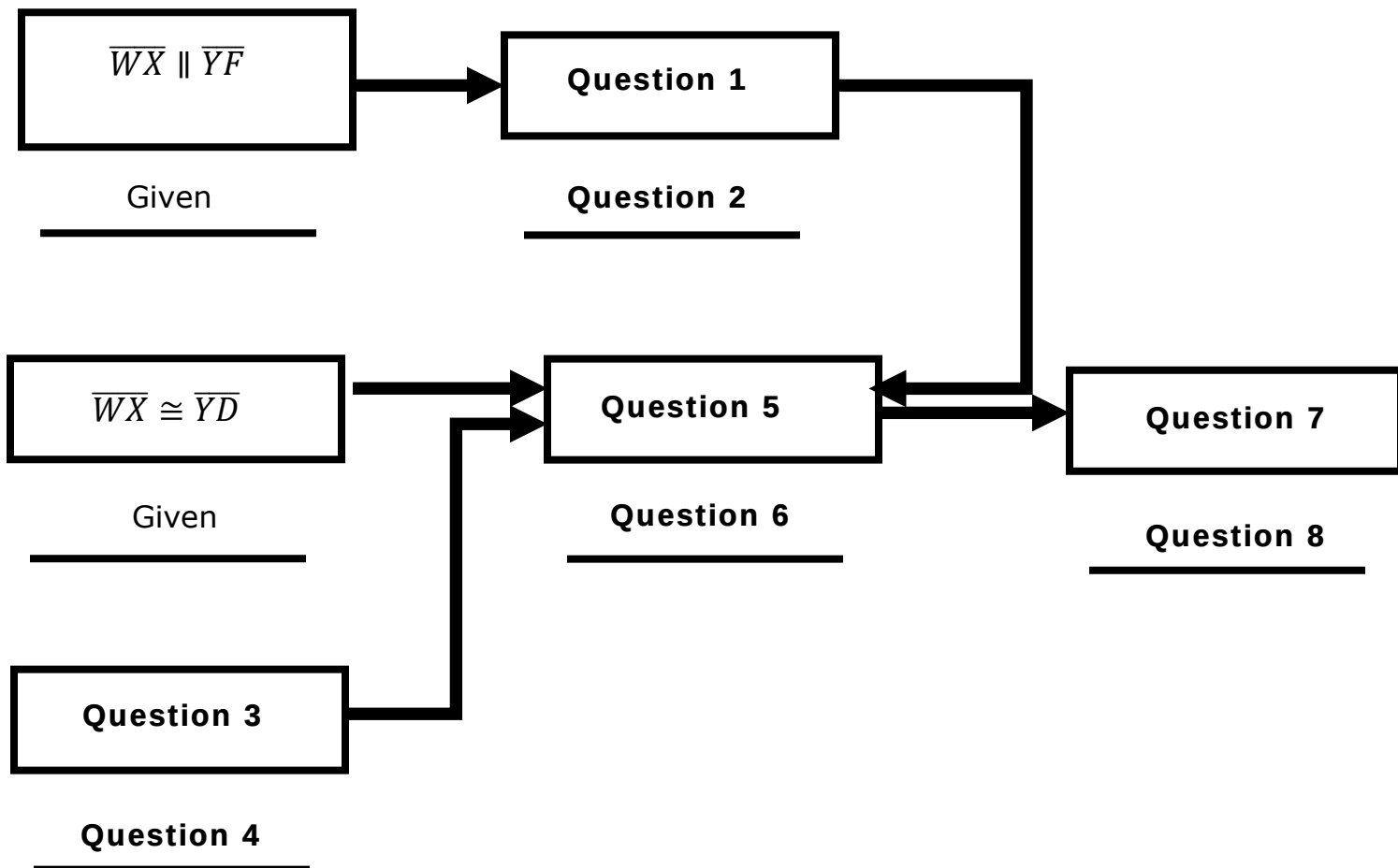
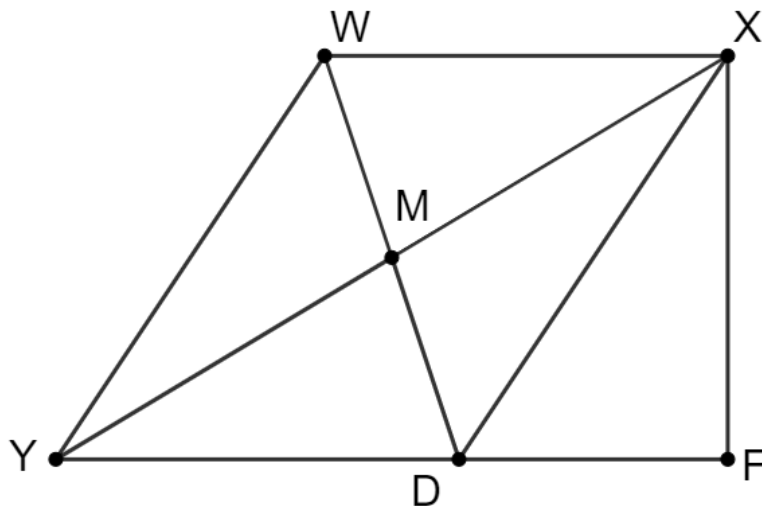
Geometry
Unit 5
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

Quadrilateral $WXYF$ is shown where
 $\overline{WX} \cong \overline{YD}$ and $\overline{WX} \parallel \overline{YF}$.

Given: $\overline{WX} \parallel \overline{YF}$, $\overline{WX} \cong \overline{YD}$
Prove: $\overline{WY} \cong \overline{XD}$

Use the given information to answer
the questions about the incomplete
proof that is shown. Each empty box
has the corresponding question
number.



1. What is the statement that belongs in the box?

$$\angle WXY \cong \angle DYX$$

2. What is the reason for the statement?

If two parallel lines are intersected by a transversal, then the alternate interior angles are congruent.

3. What is the statement that belongs in the box?

$$\overline{XY} \cong \overline{XY}$$

4. What is the reason for the statement?

Reflexive Property of Congruence

5. What is the statement that belongs in the box?

$$\triangle WXY \cong \triangle DYX$$

6. What is the reason for the statement?

SAS Congruence

7. What is the statement that belongs in the box?

$$\overline{WY} \cong \overline{XD}$$

8. What is the reason for the statement?

Corresponding Parts of Congruent Triangles are Congruent

9. Explain if it is possible to prove that $\overline{WM} \cong \overline{MD}$ using the same given statements.

Yes. Since $\overline{WX} \parallel \overline{YF}$ it can be said that $\angle WXM \cong \angle DYM$ and $\angle XWM \cong \angle YDM$ because if two parallel lines are intersected by a transversal, the alternate interior angles are congruent. Therefore $\triangle WMX \cong \triangle DYM$ by ASA $\cong$. It follows that $\overline{WM} \cong \overline{MD}$ because Corresponding Parts of Congruent Triangles are Congruent.

10. Explain if it is possible to prove $\triangle WMX \cong \triangle DMY$ using the same given statements.

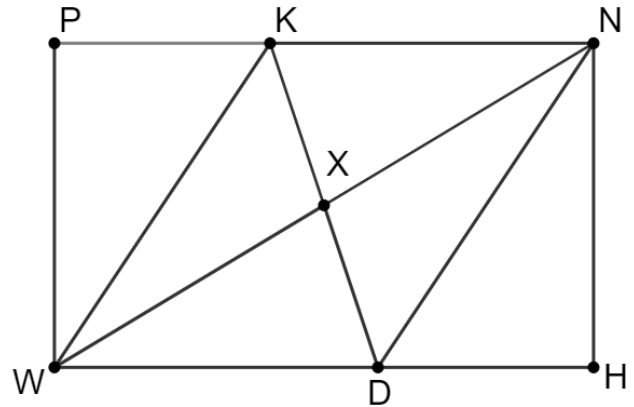
Yes. Since $\overline{WX} \parallel \overline{YF}$ it can be said that $\angle WXM \cong \angle DYM$ and $\angle XWM \cong \angle YDM$ because if two parallel lines are intersected by a transversal, the alternate interior angles are congruent. Therefore $\triangle WMX \cong \triangle DYM$ by ASA $\cong$.

Geometry
Unit 5
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

Quadrilateral $PNHW$ is shown where $\overline{KW} \parallel \overline{ND}$ and X is the midpoint of $\overline{WN}$

Given: $\overline{KW} \parallel \overline{ND}$, X is the midpoint of $\overline{WN}$
Prove: $\overline{WK} \cong \overline{DN}$



Use the given information to answer the questions about the incomplete proof that is shown.

Statement	Reason
1. $\overline{KW} \parallel \overline{ND}$	1. Given
2.	2.
3. X is the midpoint of $\overline{WN}$	3. Given
4.	4.
5.	5.
6.	6.
7. $\overline{WK} \cong \overline{DN}$	7.

1. What is the statement for line 2 of the proof?

$$\angle XKW \cong \angle XDN$$

2. What is the reason for line 2 of the proof?

If two parallel lines are intersected by a transversal, then the alternate interior angles are congruent.

3. What is the statement for line 4 of the proof?

$$\overline{WX} \cong \overline{NX}$$

4. What is the reason for line 4 of the proof?

Definition of midpoint

5. What is the statement for line 5 of the proof?

$$\angle KXW \cong \angle DXN$$

6. What is the reason for line 5 of the proof?

Vertical angles are congruent

7. What is the statement for line 6 of the proof?

$$\triangle XKW \cong \triangle XDN$$

8. What is the reason for line 6 of the proof?

AAS Congruence

9. What is the reason for line 7 of the proof?

Corresponding Parts of Congruent Triangles are Congruent

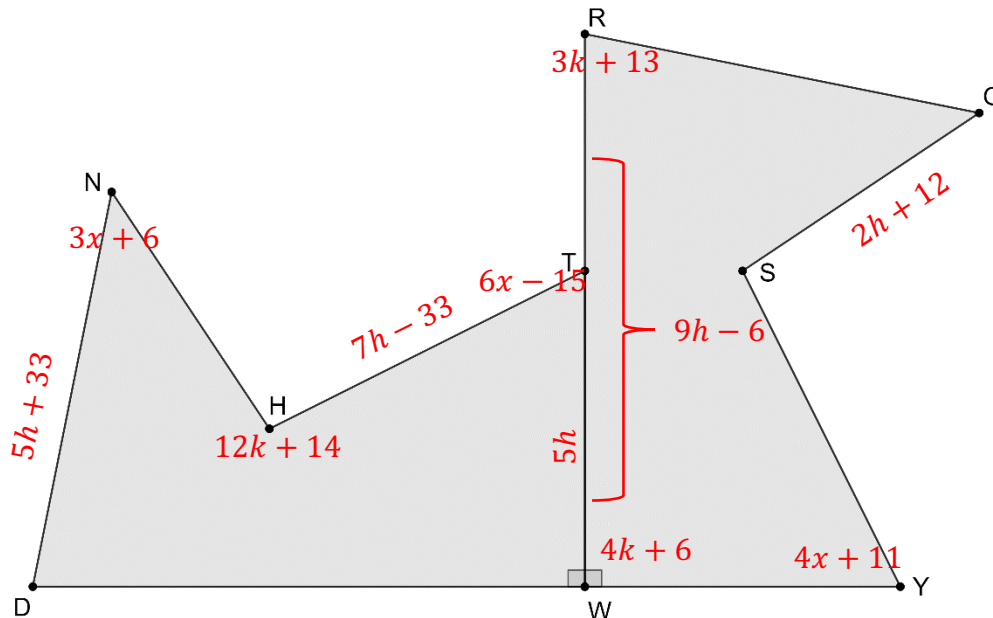
10. Explain if it is possible to prove that $\triangle WXD \cong \triangle NXK$ using the same given statements.

Yes. Using the proof shown, it can be said that $KX = DX$ by Corresponding Parts of Congruent Triangles are Congruent and $\angle KXN \cong \angle DXW$ because all vertical angles are congruent. Therefore, by SAS Congruence it can be proven that $\triangle WXD \cong \triangle NXK$.

Geometry
Unit 5
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

Congruent polygons $DNHTW$ and $RGSYW$ are shown, where $\angle DWT$ and $\angle RWY$ are right angles.



Some of the angle measures of $DNHTW$ and $RGSYW$ are shown.

$$m\angle HTW = (6x - 15)^\circ$$

$$m\angle NHT = (12k + 14)^\circ$$

$$m\angle DNH = (3x + 6)^\circ$$

$$m\angle WRG = (3k + 13)^\circ$$

$$m\angle WYS = (4x + 11)^\circ$$

$$m\angle YWR = (4k + 6)^\circ$$

1. Determine the value of k .

$$4k + 6 = 90$$

$$4k = 84$$

$$k = 21$$

2. Determine the value of x .

$$6x - 15 = 4x + 11$$

$$2x = 26$$

$$x = 13$$

Determine each angle measure.

$$\begin{aligned} 3. \quad m\angle GSY &= \underline{266^\circ} & m\angle GSY &= m\angle NHT \\ & & &= 12(21) + 14 \\ & & &= 266^\circ \end{aligned}$$

$$\begin{aligned} 4. \quad m\angle RGS &= \underline{45^\circ} & m\angle RGS &= m\angle DNH \\ & & &= 3(13) + 6 \\ & & &= 45^\circ \end{aligned}$$

$$\begin{aligned} 5. \quad m\angle WDN &= \underline{76^\circ} & m\angle WDN &= m\angle WRG \\ & & &= 3(21) + 13 \\ & & &= 76^\circ \end{aligned}$$

Some of the measures of the side lengths of $DNHTW$ and $RGSYW$ are shown.

$$ND = 5h + 33$$

$$HT = 7h - 33$$

$$RW = 9h - 6$$

$$GS = 2h + 12$$

$$TW = 5h$$

$$DW + WY = 330$$

6. Determine the value of h .

$$9h - 6 + 5h = 330$$

$$14h - 6 = 330$$

$$14h = 336$$

$$h = 24$$

7. What is the length of $\overline{HN}$?

$$HN = SG$$

$$= 2(24) + 12$$

$$= 48 + 12$$

$$= 60$$

$$HN = 60$$

9. What is the length of $\overline{RG}$?

$$RG = DN$$

$$= 5(24) + 33$$

$$= 153$$

$$RG = 153$$

8. What is the length of $\overline{SY}$?

$$SY = HT$$

$$= 7(24) - 33$$

$$= 135$$

$$SY = 135$$

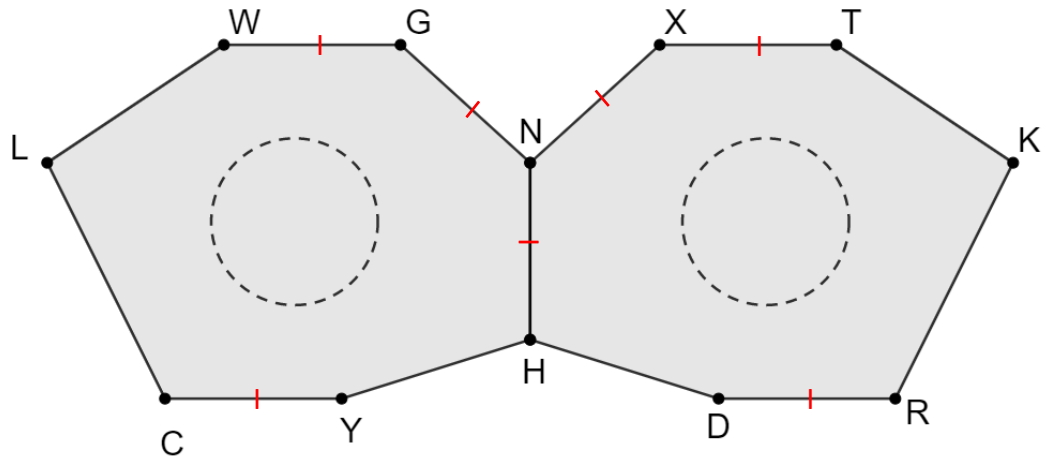
10. What is the length of $\overline{DY}$?

$$330$$

Geometry
Unit 5
Lesson 10 Practice

Name Answer Key
Date _____
Period _____

A landscape architect is designing an area in a park that will have two fountains with benches for sitting around the fountains. The two areas are congruent polygons. The architect created a diagram where congruent polygons $LWGNHYC$ and $KTXNHDR$ represent the two areas that surround the fountains.



In polygon $KTXNHDR$, $\overline{NX} \cong \overline{NH} \cong \overline{DR} \cong \overline{XT}$. The perimeter of polygon $LWGNHYC$ is 564 inches (in.)

Some of the side measures of polygons $LWGNHYC$ and $KTXNHDR$ are shown.

$$HD = 20x - 14.4$$

$$GW = 13x + 9.6$$

$$CL = 18x + 21.6$$

$$KT = 9x + 43.2$$

1. Write an equation that can be used to determine the value of x .

$$20x - 14.4 + 4(13x + 9.6) + 18x + 21.6 + 9x + 43.2 = 564$$

2. Determine the value of x .

$$20x - 14.4 + 52x + 38.4 + 18x + 21.6 + 9x + 43.2 = 564$$

$$99x + 88.8 = 564$$

$$99x = 475.2$$

$$x = 4.8$$

3. Determine the length of side HN .

$$HN = 13(4.8) + 9.6$$

$$= 72 \text{ inches}$$

4. Determine the length of side RK .

$$RK = 18(4.8) + 21.6$$

$$= 108 \text{ inches}$$

5. Determine the length of side XN .

$$XN \cong HN$$

$$= 72 \text{ inches}$$

Some of the angle measures of polygons $LWGNHYC$ and $KTXNHDR$ are shown.

$$m\angle CLW = (2k + 55)^\circ$$

$$m\angle WGN = (7k - 9)^\circ$$

$$m\angle YHN = (5k + 2)^\circ$$

$$m\angle XTK = (9k - 43)^\circ$$

$$m\angle HYC = m\angle NXT + 25^\circ$$

$$m\angle KRD = m\angle TKR + 20^\circ$$

6. Given that $m\angle XTK + m\angle TKR + m\angle KRD = 360^\circ$, determine the value of k .

$$9k - 43 + 2k + 55 + 2k + 55 + 20 = 360$$

$$13k + 87 = 360$$

$$13k = 273$$

$$k = 21$$

Determine the measure of each angle.

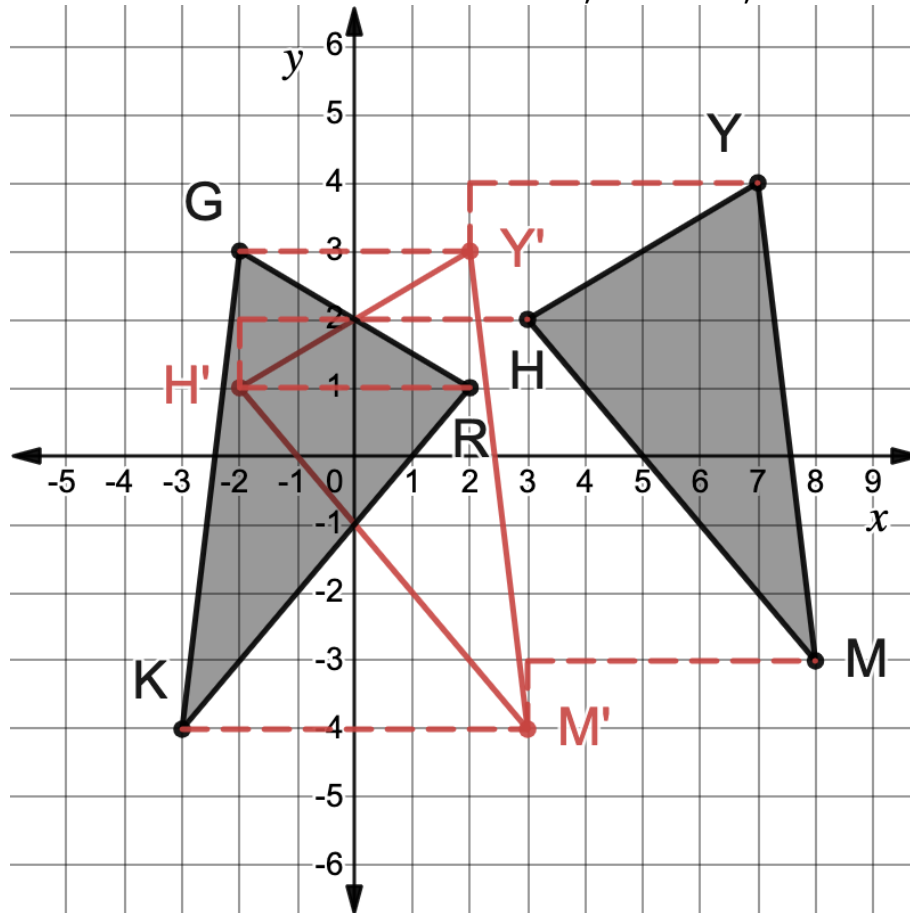
7. $m\angle NXT = \underline{138^\circ}$
 $m\angle NXT = m\angle WGN$
 $= 7(21) - 9$
 $= 138$

8. $m\angle HYC = \underline{163^\circ}$
 $m\angle HYC = m\angle NXT + 25$
 $= 138 + 25$
 $= 163$

9. $m\angle DHN = \underline{107^\circ}$
 $m\angle DHN = m\angle YHW$
 $= 5(21) + 2$
 $= 107$

10. $m\angle LWG = \underline{146^\circ}$
 $m\angle LWG = m\angle XTK$
 $= 9(21) - 43$
 $= 146$

Triangles GRK and MHY are shown. Assume $\overline{GR} \cong \overline{HY}$, $\overline{RK} \cong \overline{HM}$, and $\overline{KG} \cong \overline{YM}$.



1. Explain how to use geometric tools to verify that $\triangle HYM \cong \triangle RGK$.

Use either a compass or a ruler to show that $GR = HY$, $RK = HM$, and $KG = YM$. Then the triangles are congruent by SSS.

2. Write a sequence of transformations that will map $\triangle HYM$ onto $\triangle RGK$.

Answers may vary.

Translate $\triangle HYM$ left 6 and down 1, then reflect across the y -axis.

Show your work for questions 3 through 5 on the coordinate grid.

3. Draw $\triangle H'Y'M'$ the result of the first transformation.
4. Map the vertices of $\triangle HYM$ to $\triangle H'Y'M'$. $(x, y) \rightarrow (x - 5, y - 1)$
5. Map the vertices of $\triangle H'Y'M'$ to $\triangle RGK$. $(x, y) \rightarrow (-x, y)$

For questions 6 through 8, complete the statements.

6. The outcome of mapping the transformations of $\triangle HYM$ to $\triangle H'Y'M'$ and then from $\triangle H'Y'M'$ to $\triangle \underline{RGK}$ shows that point H' coincides with point $\underline{R}$, point Y' coincides with point $\underline{G}$, and point M' coincides with point $\underline{K}$.

7. Because the definition of congruence in terms of rigid motions preserves distance,
it can be stated that $\triangle HYM \cong \triangle \underline{RGK}$.

8. Since it was assumed that $\overline{GR} \cong \overline{HY}$, $\overline{RK} \cong \overline{HM}$, and $\overline{KG} \cong \overline{YM}$ then SSS is sufficient to show the triangles are congruent.

9. Select all of the given statements that would result in a sequence of transformations that would show SAS congruence is sufficient to show the triangles are congruent.

- ☒ $\overline{GR} \cong \overline{HY}$, $\overline{RK} \cong \overline{HM}$, $\angle GRK \cong \angle YHM$
- ☐ $\overline{GR} \cong \overline{HY}$, $\overline{KG} \cong \overline{YM}$, $\angle GRK \cong \angle YHM$
- ☒ $\overline{KG} \cong \overline{YM}$, $\overline{RK} \cong \overline{HM}$, $\angle RKG \cong \angle YMH$
- ☐ $\overline{KG} \cong \overline{YM}$, $\overline{RK} \cong \overline{HM}$, $\angle RGK \cong \angle MYH$
- ☐ $\overline{RK} \cong \overline{HM}$, $\overline{GR} \cong \overline{HY}$, $\angle RGK \cong \angle MYH$
- ☐ $\overline{RK} \cong \overline{HM}$, $\overline{KG} \cong \overline{YM}$, $\angle GRK \cong \angle YHM$

10. Select all of the given statements that would result in a sequence of transformations that would show ASA congruence is sufficient to show the triangles are congruent.

- ☐ $\angle RGK \cong \angle MYH$, $\angle GRK \cong \angle YHM$, $\overline{RK} \cong \overline{HM}$
- ☒ $\angle RGK \cong \angle MYH$, $\angle GRK \cong \angle YHM$, $\overline{GR} \cong \overline{HY}$
- ☐ $\angle GRK \cong \angle YHM$, $\angle RGK \cong \angle MYH$, $\overline{KG} \cong \overline{YM}$
- ☒ $\angle RKG \cong \angle YMH$, $\angle RGK \cong \angle MYH$, $\overline{KG} \cong \overline{YM}$
- ☒ $\angle RKG \cong \angle YMH$, $\angle GRK \cong \angle YHM$, $\overline{RK} \cong \overline{HM}$
- ☐ $\angle RKG \cong \angle YMH$, $\angle RGK \cong \angle MYH$, $\overline{GR} \cong \overline{HY}$